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Incorporating hydraulic transients into short-term operation evaluation of an integrated electricity–water system

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Abstract

Hydraulic transients occur frequently due to flow control operations in a hydraulic system, which may lead to adverse impacts on the system equipment. However, the existing researches on the power–water flow problem of the integrated electricity–water system (IEWS) are mainly based on steady-state hydraulic models, and few efforts have been spent investigating the effect of hydraulic transients on the IEWS. This paper incorporates the hydraulic transients of water distribution systems (WDSs) into the power–water flow analysis of the IEWS and proposes an extended-period quasi-steady-state IEWS model to evaluate the short-term operation of the system. The proposed model is theoretically analyzed and numerically validated based on a single-pipe WDS and a test IEWS composed of the IEEE-13 power distribution system and a 5-node WDS.

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Keywords: Integrated electricity–water system (IEWS); Quasi-steady-state model; Method of characteristics (MOC); Hydraulic transients; Power flow

1. Introduction

In recent years, there has been a worldwide effort to develop integrated energy systems (IESs) toward a more efficient, flexible and decarbonized energy system [1]. IESs explore the potential synergies among electricity, natural gas, heating and cooling systems, planning and operating various energy networks with an integrated and convergent perspective. In this context, growing interest in electricity–water interdependencies and the coordination of power distribution systems (PDSs) and water distribution systems (WDSs) is emerging, which leads to the formation and development of the concept for integrated electricity–water systems (IEWs) [2–7]. In our previous work [2], the combined power–water flow of the regional IEWS is studied, and in [3], we further propose two user-level energy-hub models and study the pressure-demand characteristics of the water load. In [4], an optimal power–water flow model of the IEWS and its distributed collaborative optimization algorithm are proposed considering network

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constraints of both the PDS and the WDS. In [5], the collaborative planning problem of the PDS and WDS is studied to improve the resilience of the overall system. Ref. [6] studies the unit commitment problem with the consideration of the regulating capacity in the WDS, viewing the WDS as a flexible resource of the power system. In [7], an optimal control model for water–energy nexus contingencies and its methodology framework for both PDS and WDS are proposed, to analyze extreme scenarios in both systems.

Power–water flow analysis is a fundamental issue in the study of IEWS planning and operation. However, among the existing IEWS studies, there are only preliminary studies in [2,3] for the power–water flow problem of the IEWS, which are mainly based on steady-state WDS models. In fact, hydraulic transients occur frequently in a hydraulic system due to sudden changes in water flow caused by flow control operations such as valve opening/closing and pump speed regulation, which propagate back and forth along the pipe through waves and may have a negative impact on the system equipment [8]. In these circumstances, the steady-state WDS models used in the present studies [2,3] may fail to provide real operation results and analyze the effects of hydraulic transients on the WDS. On the other hand, as the connection between the power system and the hydraulic system deepens, the hydraulic transients of the WDS may pose a challenge to the stable operation of the PDS. Accordingly, the hydraulic transients in the IEWS are necessary to be studied, and this is the motivation behind the current work.

To the best of our knowledge, this work is the first attempt to incorporate hydraulic transients into the power–water flow analysis and to evaluate the short-term operation of the IEWS. In the following, an extended-period quasi-steady-state model of the IEWS and a decomposed method will be discussed with both theoretical analysis and numerical studies.

2. Quasi-steady-state modeling of the IEWS

2.1. Transient-state equations of the water distribution system

2.1.1. Fundamental hydraulic transient equations

A quantitative description of hydraulic transient behaviors requires three essential equations: the continuity equation (1), the momentum equation (2), and the equation of state [9]. Since the water temperature is considered to be constant in this study, the equation of state can be ignored.

$$\frac{\partial H}{\partial t} + \frac{c^2}{gA} \frac{\partial q}{\partial x} = 0 \quad (1)$$

$$gA \frac{\partial H}{\partial x} + \frac{\partial q}{\partial t} + \frac{fq|q|}{2DA} = 0 \quad (2)$$

where t represents time; x represents pipe length; H represents hydraulic head; q denotes flow rate; c is wavespeed [8]; g is gravity acceleration; A is the cross-sectional area of pipes; D indicates pipe diameter; f is the Darcy–Weisbach friction factor [8].

However, Eqs. (1) and (2) cannot be analytically solved, so the method of characteristics (MOC) is adopted to convert the partial differential equations into ordinary differential equations that can be solved numerically using finite-difference techniques.

2.1.2. Method of characteristics

(1) Flow balance condition of nodes

The WDS consists of pipes, water sources, pumps, and water loads. Eqs. (3) and (4) describe the water flow balance at node j , which is illustrated in Fig. 1.

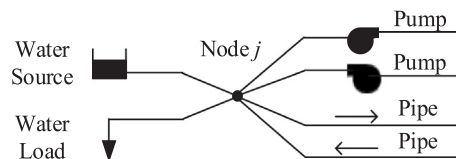


Fig. 1. Schematic diagram of the flow balance at node j .

$$q_{j,t}^S + q_{j,t}^D + q_{j,t}^{UU} + q_{j,t}^{UD} + q_{j,t}^{PU} + q_{j,t}^{PD} = 0 \quad \forall t \quad (3)$$

$$H_{j,t}^S = H_{j,t}^D = H_{j,t}^{UU} = H_{j,t}^{UD} = H_{j,t}^{PU} = H_{j,t}^{PD} \quad \forall t \quad (4)$$

where superscript S and D represent water sources and water loads, respectively; superscript UU and UD represent the upstream end and the downstream end of pumps, respectively; superscript PU and PD represent the upstream end and the downstream end of pipes, respectively.

(2) Characteristic line equation of pipes

By introducing the multiplier $\lambda = \pm c^{-1}$, Eqs. (1) and (2) are linearly combined to obtain Eq. (5) [10].

$$\left(\frac{\partial q}{\partial t} + \lambda c^2 \frac{\partial q}{\partial x} \right) + \lambda g A \left(\frac{\partial H}{\partial t} + \frac{1}{\lambda} \frac{\partial H}{\partial x} \right) + \frac{f}{2DA} q |q| = 0 \quad (5)$$

The total derivative forms of q and H are substituted into Eq. (5) to convert the equation into

$$C^+: \frac{dq}{dt} + \frac{gA}{c} \frac{dH}{dt} + \frac{f}{2DA} q |q| = 0 \quad \text{if } \frac{dx}{dt} = c \quad (6)$$

$$C^-: \frac{dq}{dt} - \frac{gA}{c} \frac{dH}{dt} + \frac{f}{2DA} q |q| = 0 \quad \text{if } \frac{dx}{dt} = -c \quad (7)$$

where Eqs. (6) and (7) represent the positive and negative characteristic lines along the two directions of the pipe, respectively. For instance, if $dx/dt=c$, Eq. (6) is valid.

By using the finite-difference technique, Eqs. (6) and (7) can be differenced as (8) and (9), respectively.

$$C^+: (q_{i,t+1} - q_{i-1,t}) + \frac{gA}{c} (H_{i,t+1} - H_{i-1,t}) + \frac{f \Delta t}{2DA} q_{i-1,t} |q_{i-1,t}| = 0 \quad \forall t \quad \text{if } \frac{dx}{dt} = c \quad (8)$$

$$C^-: (q_{i,t+1} - q_{i+1,t}) - \frac{gA}{c} (H_{i,t+1} - H_{i+1,t}) + \frac{f \Delta t}{2DA} q_{i+1,t} |q_{i+1,t}| = 0 \quad \forall t \quad \text{if } \frac{dx}{dt} = -c \quad (9)$$

where subscript i ($i = 1, 2, 3, \dots, I$) denotes the section node of the pipe, for example, $i = 1$ represents the upstream end of the pipe, and $i = I$ represents the downstream end. Δt denotes the time step.

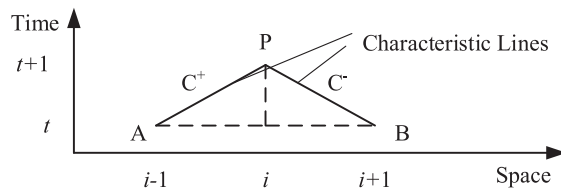


Fig. 2. Characteristic lines in space-time plane.

As shown in Fig. 2, lines C^+ and C^- indicate Eqs. (8) and (9), respectively. The flow rate (q) and hydraulic head (H) of section node P at time $(t + 1)$, space i can be obtained based on section node A (space $i - 1$) and B (space $i + 1$) at time t , respectively. Therefore, combining Eqs. (8) and (9), q and H at any time and space along pipes are available, as shown in Eqs. (10) and (11).

$$q_{i,t+1} = \frac{1}{2} \left[(q_{i-1,t} + q_{i+1,t}) + \frac{gA}{c} (H_{i-1,t} - H_{i+1,t}) - \frac{f \Delta t}{2DA} (q_{i-1,t} |q_{i-1,t}| + q_{i+1,t} |q_{i+1,t}|) \right] \quad \forall t \quad (10)$$

$$H_{i,t+1} = \frac{1}{2} \left[(H_{i-1,t} + H_{i+1,t}) + \frac{c}{gA} (q_{i-1,t} - q_{i+1,t}) - \frac{c}{gA} \frac{f \Delta t}{2DA} (q_{i-1,t} |q_{i-1,t}| - q_{i+1,t} |q_{i+1,t}|) \right] \quad \forall t \quad (11)$$

Note that to achieve a reasonable accuracy of the solution, the relationship between time step and space step specified by Courant condition [10] must be satisfied.

(3) Boundary condition of pipes

Boundary conditions, based on the hydraulic characteristics of equipment located at the pipe boundary such as water sources and pumps, are essential for the solution of the transient-state model. They determine how waves are reflected and transmitted at each pipe-ends.

For simplicity, the equations of variables Ca_{jk} , Cn_{jk} , and Cp_{jk} are introduced as follows.

$$\begin{cases} Ca_{jk} = \frac{gA}{c} \\ Cp_{jk} = q_{jk,I-1,t} + \frac{gA}{c}H_{jk,I-1,t} - \frac{f\Delta t}{2DA}q_{jk,I-1,t}|q_{jk,I-1,t}| \quad \forall t \\ Cn_{jk} = q_{jk,2,t} - \frac{gA}{c}H_{jk,2,t} - \frac{f\Delta t}{2DA}q_{jk,2,t}|q_{jk,2,t}| \quad \forall t \end{cases} \quad (12)$$

(1) Upstream-end and downstream-end conditions

The upstream end and downstream end of a pipe are determined by the water flow direction. Generally, water flows from the upstream end to the downstream end. As shown in Fig. 3, let node j represent the upstream end, then the upstream-end condition (13) (Fig. 3(a)) and the downstream-end condition (14) (Fig. 3(b)) of pipe $j - k$ can be derived from Eqs. (8) and (9), respectively.

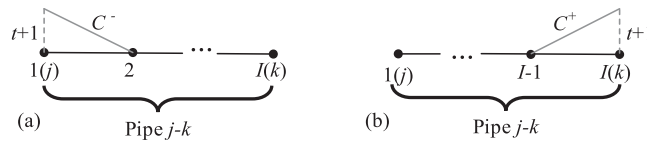


Fig. 3. (a) Upstream-end and (b) downstream-end conditions of pipe $j - k$.

$$q_{jk,1,t+1} = Cn_{jk} + Ca_{jk}H_{jk,1,t+1} \quad \forall t \quad (13)$$

$$q_{jk,I,t+1} = Cp_{jk} - Ca_{jk}H_{jk,I,t+1} \quad \forall t \quad (14)$$

(2) Water source conditions

A water source is generally abstracted as a boundary node with a constant hydraulic head, and its flow rate is equal to the upstream-end flow rate of the connected pipe.

$$H_{j,t}^S = \text{Constant}, q_{j,t}^S = q_{j,t}^{\text{PU}} \quad \forall t \quad (15)$$

(3) Pump conditions

In hydraulic transients, it is generally assumed that the steady-state characteristics of pumps can be used [11]. The flow rate $q_{jk,t}^U$ of variable-speed pump $j - k$ flows from node j to node k , and the head gain of the pump is modeled as [2]

$$\Delta H_{jk,t}^U = H_{j,t}^U - H_{k,t}^U = H_{jk}^{U0} (\omega_{jk,t}^U)^2 - c_{jk}^U (\omega_{jk,t}^U)^{2-m_{jk}^U} (q_{jk,t}^U)^{m_{jk}^U} \quad \forall t \quad (16)$$

where $\omega_{jk,t}^U$ is the pump speed ratio; H_{jk}^{U0} is the shutoff head (pump head at zero flow); c_{jk}^U and m_{jk}^U are coefficients determined by pump characteristics.

2.2. Power flow equations of the power distribution system

Electromagnetic waves propagate at the speed of light in the power system, so the transient time-scale of the power system is much smaller than that of the hydraulic system. Therefore, the steady-state model of the PDS can be adopted. For a three-phase three-wire grounded PDS, assuming that the three phases of the distributed generation are independently controllable, and all loads use a neutral grounding Y-connection method, the following three-phase power flow model can be used [3]

$$\begin{cases} P_{E,j}^\varphi(t) = V_{E,j}^\varphi(t) \sum_{k=1}^{N_e} \sum_{\psi=a,b,c} V_{E,k}^\psi(t) \{ G_{E,jk}^{\varphi\psi} \cos[\delta_{E,j}^\varphi(t) - \delta_{E,k}^\psi(t)] + B_{E,jk}^{\varphi\psi} \sin[\delta_{E,j}^\varphi(t) - \delta_{E,k}^\psi(t)] \} \\ Q_{E,j}^\varphi(t) = V_{E,j}^\varphi(t) \sum_{k=1}^{N_e} \sum_{\psi=a,b,c} V_{E,k}^\psi(t) \{ G_{E,jk}^{\varphi\psi} \sin[\delta_{E,j}^\varphi(t) - \delta_{E,k}^\psi(t)] - B_{E,jk}^{\varphi\psi} \cos[\delta_{E,j}^\varphi(t) - \delta_{E,k}^\psi(t)] \} \end{cases} \quad \forall t \quad (17)$$

where $P_{E,j}^\varphi$, $Q_{E,j}^\varphi$ are the active power injection and the reactive power injection for phase φ of node j , respectively. $V_{E,j}^\varphi$, $\delta_{E,j}^\varphi$ are the voltage amplitude and the phase angle, respectively. $G_{E,jk}^{\varphi\psi}$ and $B_{E,jk}^{\varphi\psi}$ represent the conductance and susceptance corresponding to φ , ψ phases of nodes j , k in the admittance matrix; φ , $\psi \in \{a,b,c\}$. N_e denotes the number of the PDS nodes.

2.3. Coupling between the PDS and the WDS

Electricity-driven pumps are the main coupling equipment between the PDS and the WDS. The power consumption of pump j - k is [4]

$$P_{jk,t}^U = \frac{\rho g}{\eta_{jk}^U} \Delta H_{jk,t}^U q_{jk,t}^U \quad \forall t \quad (18)$$

where ρ is water density; $\eta_{u,jk}$ denotes pump efficiency. Note that the pump power is generally viewed as a single-phase power load or a symmetric three-phase Y/ Δ load in the PDS.

2.4. Quasi-steady-state power–water flow formulations

The extended-period quasi-steady-state power–water flow equation can be formulated as follows:

$$\begin{cases} \text{PDS: (17)} \\ \text{WDS: (8) } \sim \text{ (16)} \quad \forall t \\ \text{Co: (18)} \end{cases} \quad (19)$$

This study only provides the solvability analysis of the WDS and the coupling equipment as there has been much discussion on the PDS. For a WDS with n nodes (including n_s water source nodes) and m branches (including m_u pump branches, and m_p pipe branches), the solvability analysis is provided in Table 1.

Table 1. Unknown variables and equations of the WDS and the coupling equipment.

Variable		Equation	
Description	Number	Description	Number
Flow rate	$(I-2)m_p$	Finite-difference equation of pipes (Eq. (10))	$(I-2)m_p$
	$2m_p$	Upstream-end and downstream-end conditions (Eqs. (13)–(14))	$2m_p$
	m_u	Pump head gain equation (Eq. (16))	m_u
Hydraulic head	$(I-2)m_p$	Finite-difference equation of pipes (Eq. (11))	$(I-2)m_p$
	$n-n_s$	Flow balance condition of nodes (Eq. (3))	$n-n_s$
Pump power	m_u	Pump power equation (Eq. (18))	m_u

As shown in Table 1, the number of unknown variables is consistent with the number of equations, indicating that the problem is solvable.

3. Short-term operation evaluation of the IEWS

This study solves the power–water flow of the IEWS based on the decomposed method. The flowchart of the short-term operation evaluation for the IEWS is shown in Fig. 4. Note that (1) the initial state of the WDS is assumed to be steady [9], which is obtained by EPANET [12]; (2) for the solution of the WDS model, the characteristic line equations of pipes (Eqs. (10) and (11)) need to be solved before the boundary conditions to provide the necessary variable values; (3) t is added by 1 after each iteration until the time horizon is reached, then stop the iteration and output results.

4. Case studies

The proposed model is applied to a single-pipe WDS and a test IEWS composed of the IEEE-13 PDS and a 5-node WDS. Simulations in Section 4.1 are conducted on the single-pipe WDS to introduce the propagation of hydraulic transients along the pipe. Simulations in Section 4.2 are conducted on the test IEWS to illustrate the effectiveness of the proposed model. The solution of the proposed model is based on MATLAB, which drives OpenDSS [13] to solve the power flow of the PDS, and EPANET to acquire the initial steady-state condition of the WDS.

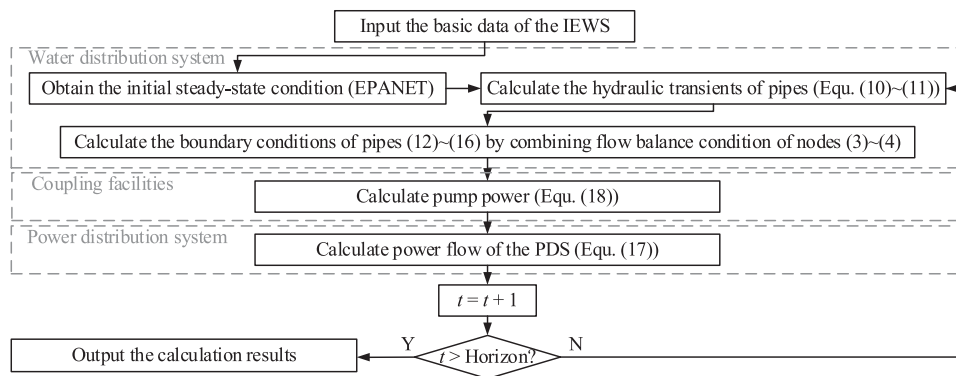


Fig. 4. Flowchart of the short-term operation evaluation for the IEWS.

4.1. Case 1: a single-pipe water distribution system

In this section, a single-pipe WDS is studied as an illustrative case to demonstrate the hydraulic transient process, as shown in Fig. 5. The pipe has a diameter of 0.35 m and a length of 4,000 m, which is divided equally into 4 sections by using the finite-difference technique to calculate the hydraulic transients. Suppose that the wavespeed c is 1000 m/s. Detailed data of Case 1 is available in [14].

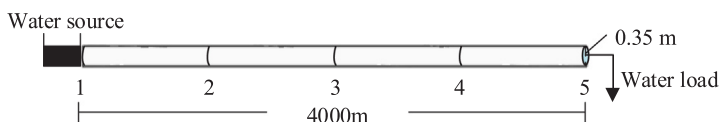


Fig. 5. The single-pipe WDS.

Suppose there is a hydraulic transient initiated by a 20% sudden reduction of the water load at 10 s, and the hydraulic transient process is shown in Fig. 6.

(1) Periodic wave analysis

As shown in Fig. 6(a), the wave propagates along the pipe toward the water source from section node 5 (abbreviated as S5), where the flow rate changes, causing a 23.57 m (64.08%) increase of the nodal hydraulic head at 10 s. After a time step (at 11 s), the wave reaches S4 and causes a 19.53% reduction in the flow rate (Fig. 6(b)) and a 22.98 m (57.33%) rise in the hydraulic head (Fig. 6(a)). During 10 s~11 s, the hydraulic condition of S5 remains unchanged due to the time needed for wave propagation. At 12 s, the wave reaches S3, and the hydraulic condition of S4 stays the same, while the hydraulic head of S5 experiences a mild increase of 1.16 m (1.92%) under the influence of the adjacent S4. Note that the flow rate of S5 remains constant because of the downstream-end boundary conditions at the pipe outlet.

When the wave reaches S1 (at 14 s), since the hydraulic head of the water source is constant, the elevated hydraulic head of the pipe distance through which the wave has traveled forms a head difference opposite to the pipe flow direction with the water source. This reverse head difference further reduces the flow rate of the water source, as shown in Fig. 6(b). By analogy, the wave will go back and forth along the pipe until it is dissipated owing to friction loss. As can be seen in Fig. 6, the fluctuation amplitudes for both the hydraulic head and flow rate attenuate until they disappear as the wave propagates.

(2) Comparison between steady-state and transient-state

Both the steady-state result calculated by EPANET and the transient-state result at the end of the simulation time for the single-pipe WDS are provided in Table 2. From Table 2, the flow rate results of steady-state are consistent with the stable values reached by the fluctuation attenuation in the hydraulic transient process, and the hydraulic head at each section node of the two states are slightly different, with a maximum of 0.16 m. This is because hydraulic

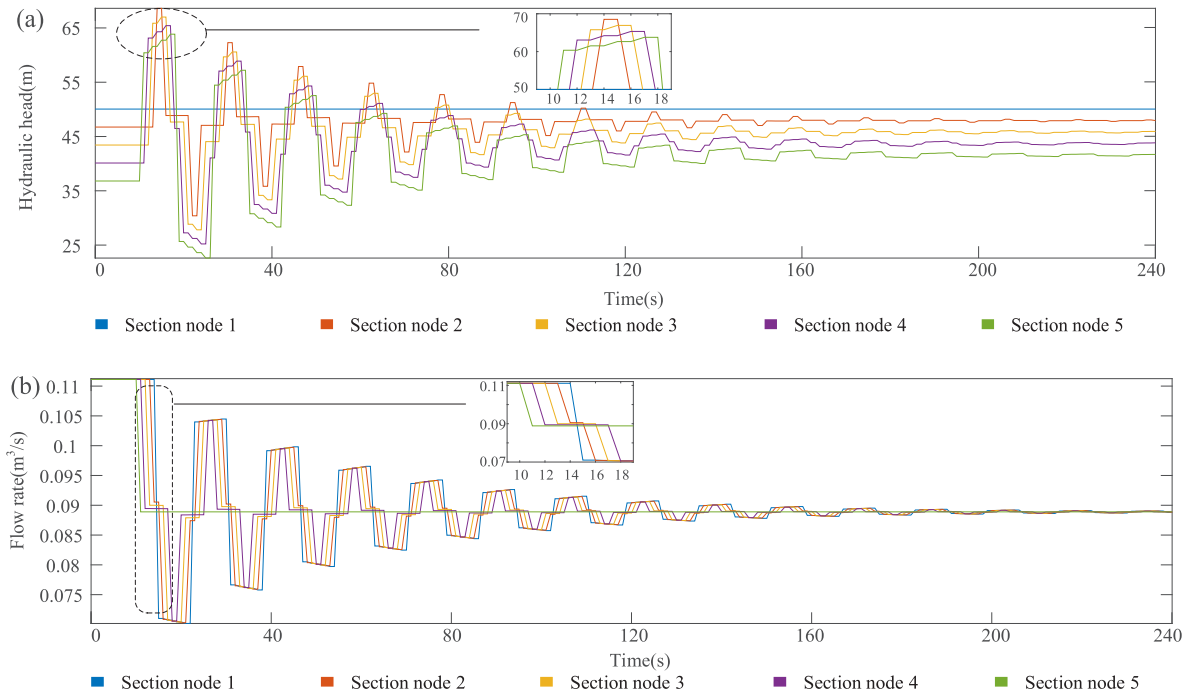


Fig. 6. The hydraulic transient process in Case 1 is shown by (a) hydraulic head and (b) flow rate at each section node of the single pipe.

Table 2. Steady-state and transient-state results in Case 1.

Section node	Flow rate (m ³ /s)			Hydraulic head (m)		
	Steady-state	Transient-state	Deviation	Steady-state	Transient-state	Deviation
1	0.089	0.089	0	50	50	0
2	0.089	0.089	0	47.88	47.9	0.02
3	0.089	0.089	0	45.76	45.88	0.12
4	0.089	0.089	0	43.63	43.79	0.16
5	0.089	0.089	0	41.51	41.67	0.16

transients are transition processes between two steady-states, and as the fluctuations attenuate, the transient result will keep converging to the next steady-state result.

4.2. Case 2: a test integrated electricity–water system

The test IEWS composed of the IEEE-13 PDS and a 5-node WDS is used to validate the proposed model, as shown in Fig. 7. Letters B and N denote PDS nodes and WDS nodes, respectively. Node B8 in the PDS supplies power to the pump in the WDS. Each pipe is 2000 m long and is divided equally into 2 sections by using the finite-difference technique to calculate the hydraulic transients. Detailed data of Case 2 is available in [14].

In this section, hydraulic transients caused by the valve control operations at water load nodes are simulated in the following two scenes.

Scene 1: Hydraulic transients are initiated by a 20% sudden reduction in the water load at N3~N5 at 10 s.

Scene 2: Hydraulic transients are initiated by a 20% sudden increase in the water load at N3~N5 at 10 s.

A 120 s short-term operation evolution of the IEWS is performed and the results of the pump operation and its connected nodal voltage in Scene 1 and Scene 2 are shown in Fig. 8 and Fig. 9, respectively. It is assumed that the power load and the pump speed remain constant during the simulation time.

(1) In Fig. 8, the blue line and the red line represent the results of transient-state and steady-state, respectively. From Fig. 8(a) and Fig. 8(b) at 10 s, it can be observed that the change of pump flow rate and pump head in

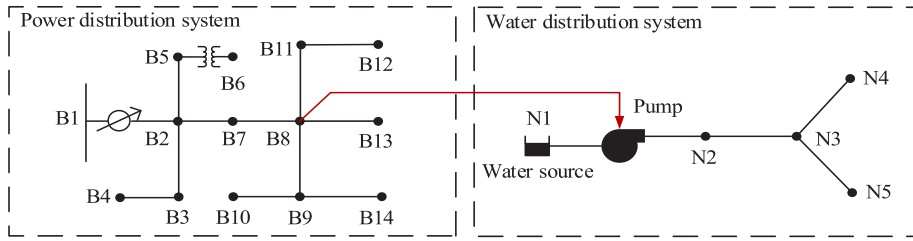


Fig. 7. The modified IEEE-13 power distribution system and 5-node water distribution system.

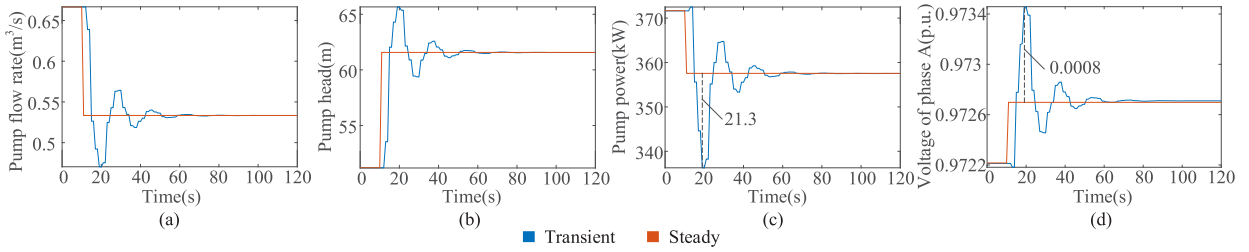


Fig. 8. Hydraulic transients of Scene 1 are shown by (a) pump flow rate, (b) pump head, (c) pump power and (d) voltage magnitude of B8.

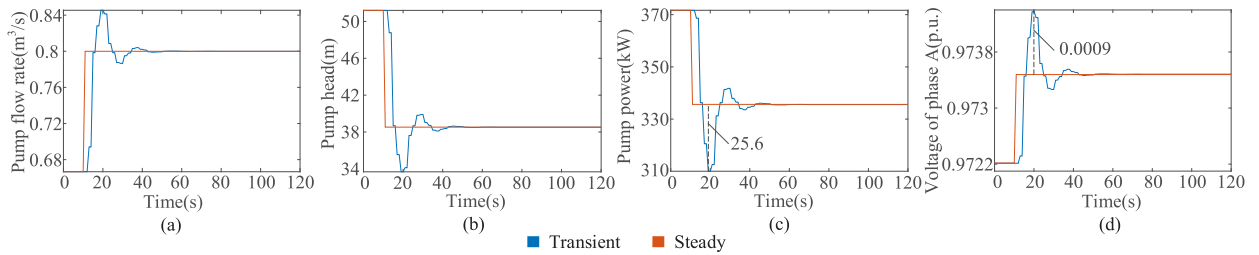


Fig. 9. Hydraulic transients of Scene 2 are shown by (a) pump flow rate, (b) pump head, (c) pump power and (d) voltage magnitude of B8.

transient-state lags behind that of the steady-state by 2 s. The reason for this is that it takes 2 s for the wave to propagate from N3, the closest among the 3 nodes (N3~N5) where the water load changes, to the pump outlet (N2). At 12 s, the pump flow rate starts to decrease and the pump head starts to increase due to the influence of the wave, as a result, the pump power in Fig. 8(c) starts to decrease and reaches a valley value of 336.3 kW at 19 s, which is 21.3 kW less than the steady-state power. Meanwhile, as shown in Fig. 8(d), the corresponding nodal voltage amplitude in the PDS starts to increase under the influence of the pump power reduction and reaches a peak value of 0.9735 at 19 s, which is an increase of 0.0008 compared to the steady-state voltage. Note that the irregularity of the fluctuation attenuation in Fig. 8 is due to the superposition effect of the waves caused by water load changes at different nodes in the WDS.

(2) In Fig. 9(a) and Fig. 9(b), the trend of the hydraulic transients in Scene 2 is opposite to that of Fig. 8(a) and Fig. 8(b) in Scene 1. At 12 s, the pump flow rate starts to increase and the pump head starts to decrease due to the influence of the wave, as a result, the pump power in Fig. 9(c) starts to decrease and reaches a valley value of 310 kW at 19 s, which is 25.6 kW less than the steady-state power. Meanwhile, the corresponding nodal voltage amplitude in the PDS in Fig. 9(d) starts to increase under the influence of the pump power reduction and reaches a peak value of 0.9744 at 19 s, which is an increase of 0.0009 compared to the steady-state voltage.

The simulation results demonstrate that the hydraulic transients caused by changes in the WDS operation state can influence the PDS operation state through coupling equipment, and during the transition to the next steady-state, the high and low extremes of power–water flow fluctuations deviate significantly from the steady-state results.

5. Conclusion

This work proposes an extended-period quasi-steady-state model of the IEWS to evaluate its short-term operation. The validity of the proposed model and method is verified numerically based on two test systems. The main feature of the proposed model is that the hydraulic transient process of the WDS is considered and incorporated into the power–water flow analysis of the IEWS.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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